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# Analysis of Benzoate 1,2-Dioxygenase Identifies Shared Electron Transfer Components With DxnA1A2 in *Rhizorhabdus wittichii* RW1

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## ABSTRACT

*Rhizorhabdus wittichii* RW1 is known for its ability to degrade polycyclic aromatic hydrocarbons, such as dibenzo-*p*-dioxin (DD) and dibenzofuran (DF). We hypothesized that the *R. wittichii* RW1 benzoate 1,2-dioxygenase shares electron transfer components with the DD/DF angular dioxygenase (DxnA1A2), similar to many aromatic hydrocarbon degrading sphingomonads. The genes encoding the benzoate oxygenase component (*benAB*) were identified in the RW1 genome sequence through homology to known benzoate oxygenases. The RW1 *benAB* genes are upstream from a putative *benD* gene encoding a *cis*-benzoate dihydrodiol dehydrogenase. Knockout of the *benA* gene resulted in a strain unable to grow on benzoate. The knockout strain could be complemented with the cloned *benABD* genes. Expression of *benAB* in *Escherichia coli* along with the *fdx3* and *redA2* genes, which encode the ferredoxin and reductase components utilized by DxnA1A2, produced a functional benzoate dioxygenase enzyme capable of converting benzoate to benzoate *cis*-dihydrodiol. Double knockout mutagenesis of the RW1 *redA1* and *redA2* reductase genes results in a mutant unable to grow on benzoate as the sole carbon source. Based on the gene knockout and heterologous expression experiments the RW1 benzoate 1,2 dioxygenase was identified and shares electron transfer components with DxnA1A2.

## 1 | Introduction

Benzoate is a pervasive aromatic hydrocarbon with both man-made and natural sources. Lignin degradation and combustion are the largest natural source for benzoate in the environment [1]. Benzoate is a common food preservative, and the largest anthropogenic source is through hydroxyl benzoate preservatives (parabens) [2]. Additionally, significant releases of benzoic acid

**Abbreviations:** *ben*AB, genes encoding the α (SWIT2634) and β (SWIT2635) subunit of the benzoate dioxygenase from *Rhizorhabdus wittichii* RW1; *benABC*, genes encoding the α (PP3161) and β (PP3162) subunit and reductase (PP3163) for the benzoate dioxygenase from *Pseudomonas putida* KT2440; BPDO, biphenyl oxygenase enzyme from *Sphingomonas yanoikuyae* B1; BphA1, salicylate oxygenase enzyme from *Sphingomonas yanoikuyae* B1; BphA3, ferredoxin enzyme from *Sphingomonas yanoikuyae* B1; BphA4, reductase enzyme from *Sphingomonas yanoikuyae* B1; CG, calcium chloride-glycerol solution; DD, compound dibenzo-*p*-dioxin (dioxin); DF, compound dibenzofuran; DXNA1A2, α and β subunit of the dioxin/dibenzofuran angular dioxygenase enzyme from *Rhizorhabdus wittichii* RW1; *fdx1*, ferredoxin gene from *Rhizorhabdus wittichii* RW1 (SWIT5088); *fdx3*, ferredoxin gene from *Rhizorhabdus wittichii* RW1 (SWIT4893); HPLC, high-performance liquid chromatography; IPTG, isopropyl β-D-1-thiogalactopyranoside; IruO, reductase enzyme from *Staphylococcus aureus*; IsdG, ring cleavage enzyme from *Staphylococcus aureus*; IsdI, ring cleavage enzyme from *Staphylococcus aureus*; MSB, mineral salts basal medium; NagAa, reductase enzyme from *Ralstonia* sp. U2; NagAb, ferredoxin enzyme from *Ralstonia* sp. U2; NagGH, naphthalene oxygenase enzyme from *Ralstonia* sp. U2; NDO, naphthalene oxygenase enzyme from *Sphingomonas yanoikuyae* B1; NEB, New England Biolabs; PhnA3, reductase enzyme from *Sphingomonas* sp. CHY-1; PhnA4, ferredoxin enzyme from *Sphingomonas* sp. CHY-1; PhnI, dioxygenase enzyme from *Sphingomonas* sp. CHY-1; PhnII, dioxygenase enzyme from *Sphingomonas* sp. CHY-1; *redA1*, reductase gene from *Rhizorhabdus wittichii* RW1 (SWIT2055); *redA2*, reductase gene from *Rhizorhabdus wittichii* RW1 (SWIT4920); S5H, salicylate oxygenase enzyme from *Ralstonia* sp. U2.

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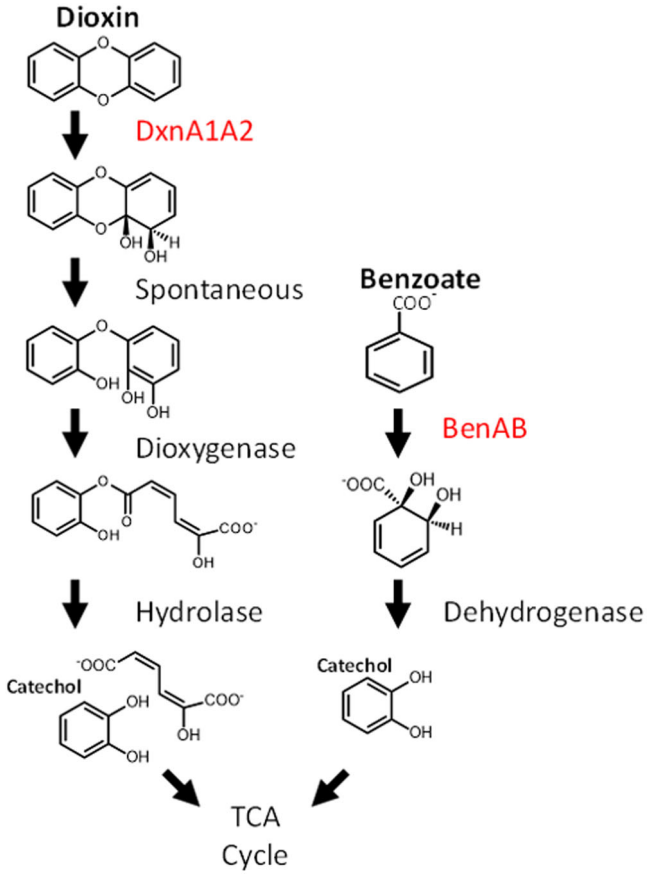



**FIGURE 1** | Degradation pathways for dioxin and benzoate by *Rhizorhabdus wittichii* RW1. The initial dioxygenases are indicated in red lettering.

and sodium benzoate into the environment primarily occur through their use in mouthwashes, dentifrices, and cosmetics [3].

Metabolism of benzoic acid by bacteria has been studied [4–6]. Benzoate degradation proceeds through catechol (Figure 1) and then either the *ortho* or *meta*-ring cleavage pathway. Key in the aerobic degradation of benzoate is a ring hydroxylating dioxygenase that catalyzes the addition of two atoms of molecular oxygen to the aromatic ring to create a *cis*-dihydrodiol intermediate [7]. Many benzoate dioxygenase oxygenase components are comprised of two oxygenase subunits. The first subunit has highly conserved histidine and cysteine residues for coordinating the Rieske type iron sulfur center (2Fe-2S), and a second subunit that plays an important role in substrate specificity [8, 9]. Dioxygenases can function in three-component systems with a short electron transfer chain involving both a ferredoxin and a reductase oxygenase [10–12], or in a two-component system with only a reductase [7, 13, 14].

*Rhizorhabdus wittichii* RW1, formerly *Sphingomonas* [15], is one of only a few strains known to fully degrade both the polycyclic aromatic hydrocarbons dibenzo-*p*-dioxin (DD) [16] and dibenzofuran (DF) [12, 17]. The first enzyme in the degradation of both DD and DF is a three-component angular dioxygenase (DxnA1A2) but, in this case, there are two interchangeable ferredoxins (Fdx1/Fdx3) and two interchangeable reductases (RedA1/RedA2) [18–20]. The reductase and ferredoxin genes have been shown to be constitutive and/or induced during growth on DD and DF [21–23].

Sphingomonads are known to encode multiple dioxygenases that share electron transfer components [11, 24–26]. In this work, we show that the RW1 DxnA1A2, and benzoate dioxygenases share common electron transfer components. We identify the RW1 *benAB* genes and show that the encoded dioxygenase enzyme component functions with the electron transfer components *redA2* and *fdx3*.

## 2 | Materials and Methods

### 2.1 | DNA Techniques and Alignments

DNA concentration and purity were measured using a Nanodrop 2000 Spectrophotometer (Thermo Fisher Scientific, Waltham, MA). Genomic DNA (gDNA) was extracted with the DNeasy UltraClean Microbial Kit (QIAGEN, Germantown, MD) and eluted in 10 mM tris-EDTA buffer. Plasmid DNA was purified using the Macherey-Nagel NucleoSpin plasmid kit (Bethlehem, PA) following the manufacturers' instructions. PCR amplifications were performed using the Phusion high-fidelity kit (New England Biolabs (NEB), Ipswich, MA), and all PCR-generated fragments used in downstream applications were sequenced to ensure no base changes occurred. Restriction enzymes and DNA ligations were carried out as recommended by the supplier (NEB). PCR products and restriction fragments were purified using the GeneClean III kit (MP Biomedicals, Santa Ana, CA) according to the manufacturer's instructions. Following amplification with the Phusion enzyme, 5 μL of the PCR product was mixed with 5 μL Taq 2X polymerase mix (NEB) to adenylate fragment ends for TA cloning into the pGEM-T Easy vector system (Promega, Madison, WI). The final constructs were verified by size using a 0.8% agarose gel. DNA sequencing was performed by Azenta (South Plainfield, NJ). Similar nucleotide sequences were identified using BlastN against the GenBank database, and the full benA gene (from start to stop codon) was extracted from selected sequences. These sequences were aligned using ClustalW in the Lasergene MegAlign software (DNAStar, Madison, WI). The dendrogram was created using FigTree software (http://tree.bio.ed.ac.uk/software/figtree/). The GenBank locus tags (SWIT numbers) and protein ID numbers for the genes analyzed in *R. wittichii* RW1 are: SWIT0053/ABQ66426, SWIT0285/ABQ66656, SWIT0363/ABQ66734, SWIT0998/ABQ67365, SWIT1843/ABQ68203, SWIT2055/ABQ68414 (*redA1*), SWIT2634/ABQ68992 (*benA*), SWIT2635/ABQ68993 (*benB*), SWIT2636/ABQ68994 (*benD*), SWIT4893/ABQ71515 (*fdx3*), SWIT4920/ABQ71540 (*redA2*), and SWIT5088/ABQ71698 (*fdx1*). The GenBank locust tags (PP numbers) and protein ID numbers for the genes used as positive controls in *Pseudomonas putida* KT2440 are: PP3161/AAN68769 (KT_*benA*), PP3162/AAN68770 (KT_*benB*), and PP3163/AAN68771 (KT_*benC*). The GenBank gene names and protein ID numbers for the benzoate dioxygenase genes in *Sphingomonas yanoikuyae* B1 are: *xylX*/ABM79785 (α subunit) and *xylY*/ABM79786 (β subunit). The amino acid alignment was visualized using the Constraint-based Multiple Alignment Tool (COBALT) from NCBI (https://www.ncbi.nlm.nih.gov/tools/cobalt).

## 2.2 | Cells, Solutions, and Assays

*R. wittichii* RW1 was cultivated in mineral salts basal (MSB) medium [27, 28], supplemented with defined carbon sources and antibiotic selection, as described in the text. To prepare competent *Escherichia coli* cells, a calcium chloride–glycerol (CG) solution [29] was utilized for both *Escherichia coli* DH5α and *E. coli* BL21 DE3. For resting cell assays and subsequent high-performance liquid chromatography (HPLC) analysis of product formation, a 50 mM sodium/potassium phosphate buffer was employed [27], supplemented with 20 mM glucose and 1 mM benzoate. *E. coli* DH5α facilitated plasmid construct generation and screening, whereas *E. coli* BL21 DE3 served as the host for heterologous protein expression. *E. coli* HB101 harboring pRK2013 was employed to mediate tri-parental mating procedures. *R. wittichii* RW1 possesses benzoate dioxygenase and associated electron transfer components, which remain under active investigation. Additionally, *Pseudomonas putida* KT2440 was incorporated as a reference strain, utilizing its *benABC* gene cluster as a positive control for cis-dihydrodiol formation. Table 1 provides a comprehensive overview of all bacterial strains and plasmids used in this study.

## 2.3 | Heterologous Expression Constructs

The expression vector pGGA was used for this study. The 1.86 kb *benAB* genes were amplified with the primers pGEMT_BenAB_F (added NotI site) and pGEMT_BenAB_R (added BbsI and EarI sites) and were inserted into the pGEM-T Easy vector. Table 2 lists all the primers used in this study. After confirmation of the correct sequence, the *benAB* fragment was moved to pGGA via NotI and BbsI (NEB) to create pGGA_*benAB*.

The 1.2 kb *redA2* fragment was amplified with the primers pGEMT_RedA2_F (added EarI site) and pGEMT_RedA2_R (added EarI site) and was inserted into the pGEM-T Easy vector. The *redA2* fragment was moved to pGGA_*benAB* via EarI (NEB) to create pGGA_*benAB_redA2*.

The ferredoxin genes were amplified with the primers pGEMT_Fdx3_F (added EarI site), pGEMT_Fdx3_R (added BamHI site), pGEMT_0053_F (added EarI site), pGEMT_0053_R (added BamHI site), pGEMT_0285_F (added EarI site), pGEMT_0285_R (added BamHI site), pGEMT_0363_F (added EarI site), pGEMT_0363_R (added BamHI site) and were inserted into pGEM-T Easy vector. The ferredoxins were all moved to pGGA_*benAB_redA2* using their respective restriction enzymes (NEB) to create the final three component constructs.

The *benABC* genes were amplified from *P. putida* KT2440 with the primers pGEMT_BenDiox_F (added NotI site) and pGEMT_BenDiox_R (added BsaI site) and were inserted into the pGEM-T Easy vector. The *benABC* genes were moved to pGGA using their respective restriction enzymes (NEB) to create pGGA_*BenDiox*. *E. coli* DH5α containing the pGGA constructs were plated on Luria-Bertani agar (LB) supplemented with 33 μg/mL chloramphenicol.

## 2.4 | Knockout Mutagenesis of *benAB* and Complementation

The low copy vector pRK415 facilitated gene knockout experiments, as described previously [11, 24, 34–36]. The RW1 *benAB* genes, including upstream and downstream flanking regions, were amplified into a 3.8 kb fragment using pGEMT_BenAB_KO_F (HindIII) and pGEMT_BenAB_KO_R (XbaI), then cloned into pGEM-T Easy. The *benAB*_KO construct was transferred into pRK415 via HindIII and XbaI (NEB), generating pRK415_*benAB*_KO. Constructs were introduced into *E. coli* DH5α and screened on LB with 12 μg/mL tetracycline and 20 μg/mL X-Gal.

To confer gentamycin resistance, the p34S_GM-derived resistance cassette was inserted into pRK415_*benAB*_KO using SalI and XhoI (NEB), yielding pRK415_*benAB*_KO_Gm. *E. coli* DH5α transformants were screened on LB containing 12 μg/mL tetracycline and 50 μg/mL gentamycin. For complementation, the *benABD* genes were amplified with pGEMT_BenABD_F (KpnI) and pGEMT_BenABD_R (EcoRI), cloned into pGEM-T Easy, and transferred into pRK415 via KpnI and EcoRI (NEB), forming pRK415_*benABD*. Transformants were screened on LB with 12 μg/mL tetracycline and 20 μg/mL X-Gal.

Knockout mutagenesis involved a tri-parental mating of *E. coli* DH5α(pRK415_*benAB*_KO_Gm), *E. coli* JM109(pRK2013), and *R. wittichii* RW1 at a 1:1:2 ratio, incubated on LB overnight at 30°C. Colonies were selected on MSB with 5 mM phenylalanine and 12 μg/mL tetracycline and re-streaked twice to remove *E. coli* contaminants. Serial transfers in LB were performed daily, with initial selection on 12 μg/mL tetracycline, followed by screening on 50 μg/mL gentamycin after ten transfers. Loss of pRK415 was confirmed by tetracycline screening, and knockouts (RW1Δ*benA*) were validated via Taq PCR (NEB). Complementation was performed by mating *E. coli* DH5α (pRK415_*benABD*), *E. coli* JM109(pRK2013), and *R. wittichii* RW1Δ*benA* at a 1:1:2 ratio.

## 2.5 | Growth Curve and HPLC

For growth curves, overnight LB cultures were subcultured in MSB with 5 mM benzoate (initial OD600 = 0.05), incubated at 30°C with shaking at 180 rpm, and sampled at 0, 5, 10, 15, and 25 h. Expression constructs, including BL21 DE3(pGGA_*benAB_redA2_fdx3*), were cultured in LB with 33 μg/mL chloramphenicol and 1 mM IPTG until OD600 reached 0.8. Cells were pelleted (10,000 g, 8 min), washed twice with phosphate buffer, and resuspended in 25 mL phosphate buffer supplemented with 1 mM benzoate and 20 mM glucose. Cultures were incubated at 30°C with shaking at 180 rpm, and supernatants were collected after centrifugation (10,000 g, 5 min). Cell-free extracts from *E. coli* BL21 DE3 containing only pRK415 served as negative controls. Samples were taken at 0, 4-, 8-, 12-, 16-, 20- and 24-h time points.

Samples were analyzed by HPLC with a Waters Symmetry C8, 4.6 × 250 mm with 5 μm packing column (Waters, Milford, MA) with detection at 254 nm. The mobile phase was 1.0% acetic acid in water with a gradient of 0%–100% of 1.0% acetic acid in methanol over 90 min. BL21 DE3(pGGA_KT_*benABC*) was

**TABLE 1** | List of strains and plasmids for this study.

| Strain/plasmid | Description | Reference/source |
|---|---|---|
| RW1 | *Rhizorhabdus wittichii* RW1 wild type strain | DSMZ (DSM 6014) |
| RW1Δ*benA* | RW1 with gentamycin resistance cassette inserted into *benA* | This study |
| RW1Δ*fdx1* | RW1 with deletion of *fdx1* | [30] |
| RW1Δ*fdx3* | RW1 with deletion of *fdx3* | [30] |
| RW1Δ*fdx1*Δ*fdx3* | RW1 with deletion of both *fdx1* and *fdx3* | [30] |
| RW1Δ*redA1* | RW1 with deletion of *redA1* | [30] |
| RW1Δ*redA2* | RW1 with deletion of *redA2* | [30] |
| RW1Δ*redA1*Δ*redA2* | RW1 with deletion of both *redA1* and *redA2* | [30] |
| KT2440 | *Pseudomonas putida* KT2440 used for generate the PCR fragment KT_*benABC* | DSMZ (DSM 6125) |
| BL21 DE3 | *Escherichia coli* BL21 DE3 used for heterologous expression | Invitrogen |
| DH5α | *Escherichia coli* DH5α used for transferring constructs | Invitrogen |
| JM109 | *Escherichia coli* JM109 contains the helper plasmid needed for conjugation experiments | Invitrogen |
| pGEM-T Easy | TA cloning vector | Promega |
| pGEMT_*BenDiox* | pGEM-T Easy containing *benABC* | This study |
| pGEMT_*benAB* | pGEM-T Easy containing *benAB* | This study |
| pGEMT_*benAB*_KO | pGEM-T Easy containing *benA* with flanking regions | This study |
| pGEMT_*benABD* | pGEM-T Easy containing *benABD* | This study |
| pGEMT_*redA2* | pGEM-T Easy containing *redA2* | This study |
| pGEMT_*fdx3* | pGEM-T Easy containing *fdx3* | This study |
| pGEMT_0053 | pGEM-T Easy containing SWIT 0053 | This study |
| pGEMT_0285 | pGEM-T Easy containing SWIT 0285 | This study |
| pGEMT_0363 | pGEM-T Easy containing SWIT 0363 | This study |
| pGEMT_0998 | pGEM-T Easy containing SWIT 0998 | This study |
| pGEMT_1843 | pGEM-T Easy containing SWIT 1843 | This study |
| pGGA | pGGA cloning vector | NEB |
| pGGA_*benAB* | pGGA containing *benAB* | This study |
| pGGA_*BenDiox* | pGGA containing *benABC* | This study |
| pGGA_*benAB_redA2* | pGGA containing *benAB* and *redA2* | This study |
| pGGA_*benAB_fdx3* | pGGA containing *benAB* and *fdx3* | This study |
| pGGA_*benAB_redA2*_fdx3 | pGGA containing *benAB*, *redA2* and *fdx3* | This study |
| pGGA_*benAB_redA2*_0053 | pGGA containing *benAB*, *redA2* and SWIT_0053 | This study |
| pGGA_*benAB_redA2*_0285 | pGGA containing *benAB*, *redA2* and SWIT_0285 | This study |
| pGGA_*benAB_redA2*_0363 | pGGA containing *benAB*, *redA2* and SWIT 0363 | This study |
| pGGA_*benAB_redA2*_0998 | pGGA containing *benAB*, *redA2* and SWIT 0998 | This study |
| pGGA_*benAB_redA2*_1843 | pGGA containing *benAB*, *redA2*, and SWIT 1843 | This study |
| pRK415 | Unstable broad host range cloning vector | [31] |
| pRK415_*benAB*_KO | pRK415 containing *benA* with flanking regions | This study |
| pRK415_*benAB*_KO_Gm | pRK415_BenA_KO with gentamycin resistance cassette cloned into *benAB* | This study |
| pRK415_*benABD* | pRK415 containing *benABD* | This Study |
| pRK415_*redA1* | pRK415 containing *redA1* | [30] |
| pRK415_*redA2* | pRK415 containing *redA2* | [30] |
| pRK415_*redA1_redA2* | pRK415 containing *redA1* and *redA2* | [30] |
| pRK415_*fdx1* | pRK415 containing *fdx1* | [30] |
| pRK415_*fdx3* | pRK415 containing *fdx3* | [30] |

(Continues)

**TABLE 1** | (Continued)

| Strain/plasmid | Description | Reference/source |
|---|---|---|
| pRK415_*fdx1_fdx3* | pRK415 containing *fdx1* and *fdx3* | [30] |
| pRK2013 | Helper plasmid for conjugation experiments | [32] |
| p34S-Gm | Source of gentamycin resistance cassette | [33] |

**TABLE 2** | List of PCR primers for this study.

| Initial construct | PCR primer | PCR product |
|---|---|---|
| pGEMT_BenAB | GGGCGGCCGCGCAGAGGAGAGCTTCCATGA<br>GGGAAGACTTGGTAGAAGAGCTCTTCTTCGAACATGTCCGGCTCCT | *benAB* for heterologous expression |
| pGEMT_BenAB_KO | CCAAGCTTATGCGGTGGCTCATGATGAT<br>CCTCTAGAGGTTGGCATAGTAGATCAGC | *benA* with flanking regions for knockout construct |
| pGEMT_BenABD | CCGGTACCCGAGATCGGATTGATCTTTGC<br>CCGAATTCCATATCCTCCGTCTCTGTCG | *benABD* for complementation |
| pGEMT_BenDiox | GCGGCCGGAGACCCTGGCCATGTCCC<br>GGGGAGCGAGACCGCAACTCTCTTGCTCAAAGC | *benABC* for heterologous expression |
| pGEMT_RedA2 | GGTATGGAAGAGGCTAAGGAGGGCAGGTTCTGG<br>GGGGTAGAAGAGGGCGTCTCCCTGCTCCAATAG | *redA2* for heterologous expression |
| pGEMT_Fdx3 | GGCACGGAAGAGGATGGGAGATAATAGTGATGC<br>CCGGATCCCTTTGGAGGATCGTTAAGTTC | *fdx3* for heterologous expression |
| pGEMT_0053 | GGCACGGAAGAGGGAGATAATAGTGATGACCAGGGTCGTGTTCATC<br>CCGGATCCGAATTTGCTGGTCCGACCGA | SWIT 0053 for heterologous expression |
| pGEMT_0285 | GGCACGGAAGAGCGCCCTAGGAAGACATGATG<br>CCGGATCCGTATCCGATCACATCCTGAAC | SWIT 0285 for heterologous expression |
| pGEMT_0363 | GGCACGGAAGAGGTTTCGGAGACTGACCGCATG<br>CCGGATCCCTCGGCCATGGCTCAATCCT | SWIT 0363 for heterologous expression |
| pGEMT_0998 | GGCACGGAAGAGGATCAGCTATCTCGATCAGG<br>CCGGATCCTCAGCACGACCAGCCCGTCC | SWIT 0998 for heterologous expression |
| pGEMT_1843 | GGCACGGAAGAGCCTCCCATGCCGATGATCAC<br>CCGGATCCGTGCCGCCTCAGCCCTGACG | SWIT 1843 for heterologous expression |

used as a positive control to confirm the retention time of benzoate *cis*-dihydrodiol. Under these conditions, benzoate and benzoate *cis*-dihydrodiol had retention times of 26.8 and 7.4 min, respectively.

## 3 | Results

### 3.1 | Dibenzo-*p*-Dioxin and Benzoate Degradation Are Linked

While investigating electron transport components utilized in DD degradation it was discovered that the two interchangeable reductases are not only necessary for DD degradation, but benzoate degradation as well [30] (Figure 2). Single knockouts of either *redA1* or *redA2* did not affect growth on benzoate. A double knockout of both reductases resulted in a strain not able to grow on benzoate and was complemented by addition of either *redA1* or *redA2* (Figure 2). The two interchangeable ferredoxins in DD degradation were also tested through knockout mutagenesis but double knockout of both *fdx1* and *fdx3* did not affect growth on benzoate (Figure 2).

### 3.2 | Identifying Candidates for the Benzoate Dioxygenase in *R. wittichii* RW1

To identify candidates for the benzoate dioxygenase in *R. wittichii* RW1, gene homology to known benzoate oxygenases in *P. putida* KT2440 and *S. yanoikuyae* B1 was conducted and compared to 12 known dioxygenase α subunits from RW1. SWIT_2634 was identified as the alpha subunit of benzoate dioxygenase in RW1 (*benA*) (Figure 3). Both protein subunits of the benzoate dioxygenase (BenAB) in RW1 are evolutionarily distinct from the KT2440 and B1 strains, with the BenA subunit having 60% amino acid similarity to the KT2440 enzyme and 80% amino acid

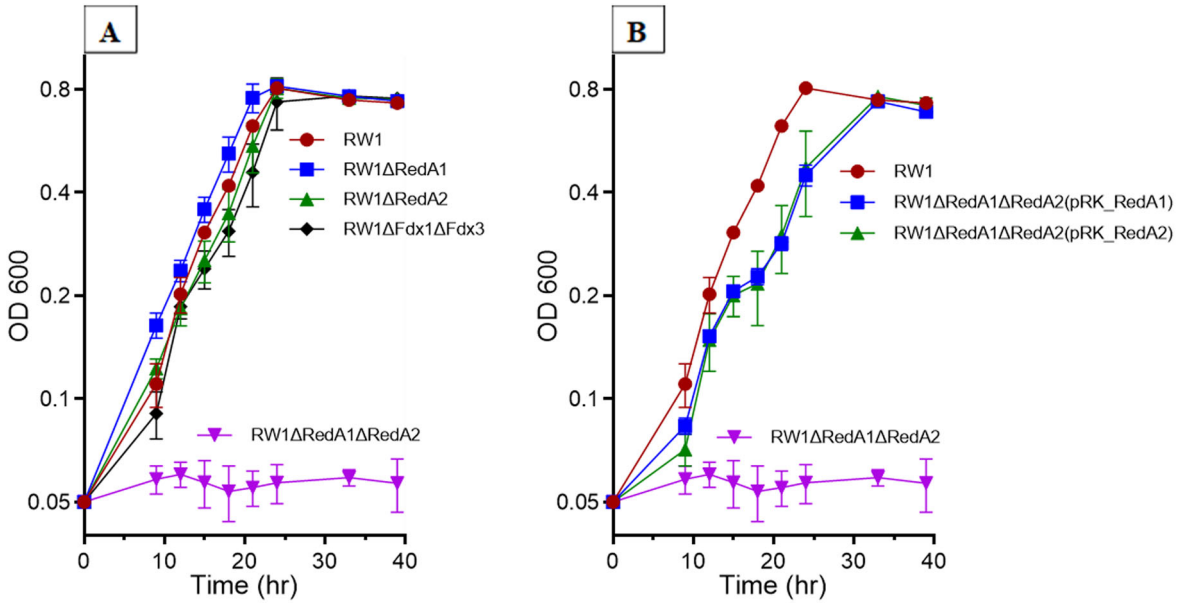



**FIGURE 2** | (A) RW1 electron transport mutants on MSB supplemented with 5 mM benzoate. (B) Complement of the RedA1/RedA2 mutant on MSB supplemented with 5 mM benzoate. Data are averages from triplicates, and the error bars indicate their standard deviations.

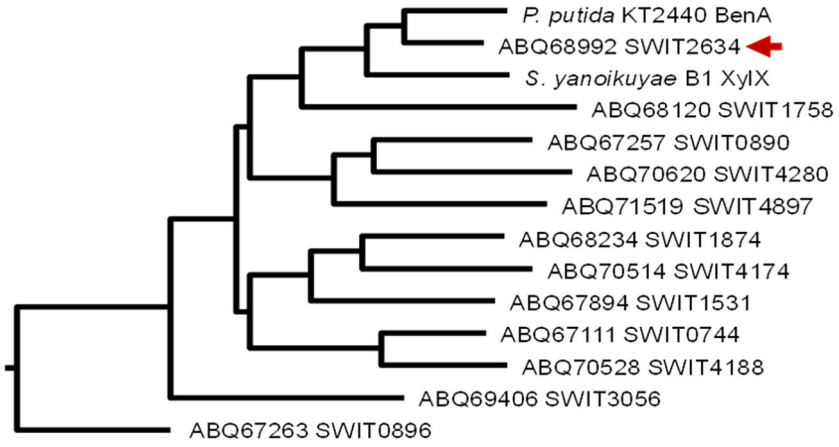



**FIGURE 3** | Dendrogram comparing the *Pseudomonas putida* KT2440 *benA* and *Sphingomonas yanoikuyae* B1 *xylX* genes to oxygenase genes in *Rhizorhabdus wittichii* RW1. The red arrow indicates the *benA* gene in RW1.

similarity to the B1 enzyme, while the BenB subunit has 61% similarity to the KT2440 enzyme and 68% similarity to the B1 enzyme (Figure 4).

### 3.3 | Knockout Mutagenesis of *benAB*

To confirm that *benAB* was the only benzoate dioxygenase in RW1 knockout mutagenesis was performed. Knockout mutagenesis of SWIT_2634 resulted in a mutant (RW1Δ*benA*) unable to grow on benzoate as the sole carbon source (Figure 5). An attempt was made to complement RW1Δ*benA* with pRK_*benA*, however the gene for the α subunit failed to restore growth on benzoate (Figure 5). To remedy this, all three genes in the benzoate operon (*benABD*) were cloned into pRK415 for complementation experiments. The RW1Δ*benA*(pRK_*benABD*) strain restored the ability to grow on benzoate (Figure 5). This result suggests that the *benA* knockout mutation caused a polar effect on downstream genes.

### 3.4 | Heterologous Expression of *benAB* in *E. Coli* BL21 DE3

The positive control BL21(pGGA_*BenDiox*) containing the *P. putida* KT2440 *benABC* genes for benzoate dioxygenase converted 100% (1 mM) benzoate to benzoate dihydrodiol in 24 h (Figure 6). BL21(pGGA_*benAB*) was not able to produce any dihydrodiol intermediate in 24 h (Figure 6). Since reductase knockouts implicated *redA1* and *redA2*, constructs incorporating *redA2* were designed (pGGA*benABredA2*). The addition of redA2 did not result in the conversion of benzoate to a dihydrodiol (Figure 6). It was hypothesized that BenAB functions as a three component system, and so *fdx3* was added




**FIGURE 4** | Amino acid sequence alignment. (A) The α subunit; (B) the β subunit. The sequences include BenAB from *Rhizorhabdus wittichii* RW1 (top sequence), BenAB from *Pseudomonas putida* KT2440 (middle sequence) and XylXY from *Sphingomonas yanoikuyae* B1 (bottom sequence). Red lettering indicates conserved amino acid regions.

(pGGA_*benAB_redA2_fdx3*) to see if the BenAB dioxygenase can utilize the exact same three component system as the DD dioxygenase. BL21(pGGA_*benAB_redA2_fdx3*) was able to completely convert 1 mM benzoate to the dihydrodiol in a 24 h time period (Figure 7).

Since knockouts of *fdx1* and *fdx3* did not abolish growth of RW1 on benzoate, it was hypothesized that there was a third ferredoxin which could support the three component system. There was motive to test all the known ferredoxins in the RW1 genome which fall into the same class of ferredoxin as Fdx3; ferredoxins whose iron sulfur cluster active site is held into place by 4 cysteine amino acid residues. The genes SWIT 0053, SWIT 0285, SWIT 0363, SWIT 0998, and SWIT 1843 fell into this class of enzyme. The other candidate ferredoxins were moved to the pGGA_*benAB_redA2* expression construct to see if they could produce the dihydrodiol. SWIT 0285, SWIT 0998 and SWIT 1843 did not have any activity while addition of SWIT 0053 resulted in an average of 5% conversion and SWIT 0363 resulted in an average of 12.4% conversion of benzoate to the *cis*-dihydrodiol over a 24 h time period (Figure 6).

## 4 | Discussion

In conclusion, the *benAB* genes (SWIT 2634 and SWIT 2635) from *Rhizorhabdus witichii* RW1 encode the benzoate dioxygenase subunits. Either the RedA1 or the RedA2 reductase can function as the reductase component of the three-component dioxygenase enzyme. Multiple different ferredoxins, including Fdx3, can function as the ferredoxin component of the dioxygenase enzyme. This has implications for future remediation efforts utilizing *R. wittichii* RW1 to clean up DD and DF since the DD, DF, and benzoate degradation pathways compete for the same electron transfer components to complete their enzymatic reactions. Since benzoate is a highly pervasive aromatic hydrocarbon [1–3], there is a high likelihood of co-contamination in sites polluted by DD and DF.

The ability to share electron transport components by different oxygenase systems is not unique to RW1. *Sphingomonas yanoikuyae* B1 utilizes the same ferredoxin (BphA3) and reductase (BphA4) for its salicylate (BphA1c), naphthalene (NDO) and biphenyl (BPDO) oxygenase enzymes [11, 24]. *Ralstonia* sp. U2 utilizes the same reductase (NagAa) and ferredoxin (NagAb) for

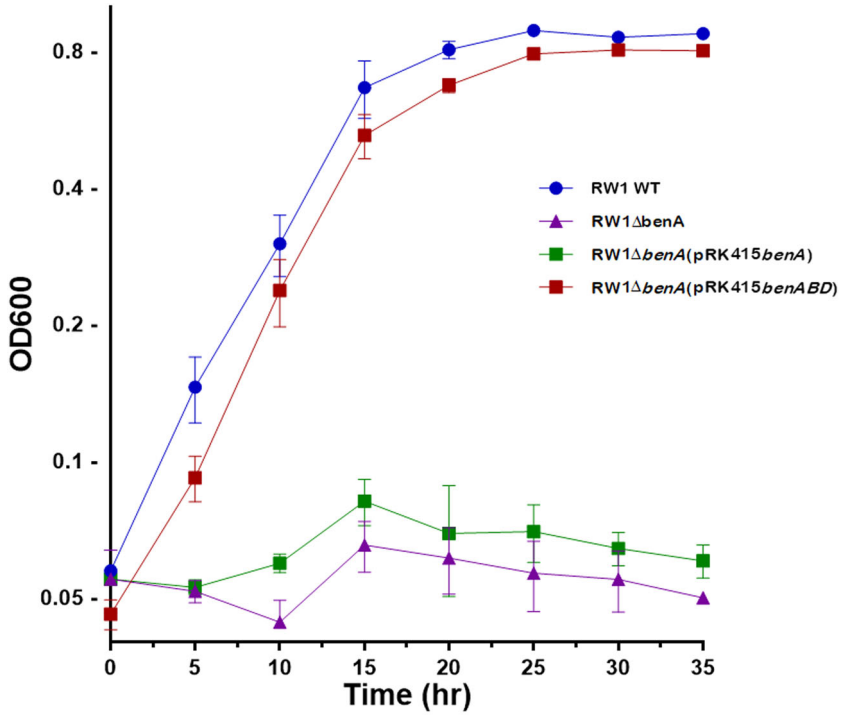



**FIGURE 5** | Growth curve of *benAB* mutants on MSB supplemented with 5 mM benzoate. Data are averages from triplicates, and the error bars indicate their standard deviations.

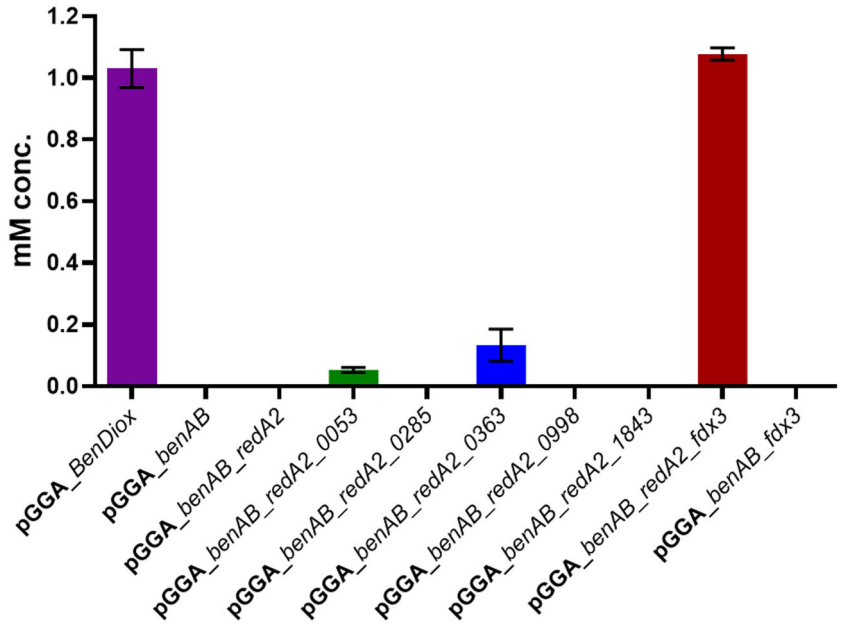



**FIGURE 6** | Total *cis*-dihydrodiol production by *Escherichia coli* BL21 DE3 using the *benAB* expression constructs with the various electron transfer components. Assays were performed for 24 h, data are averages from triplicates, and the error bars indicate their standard deviations.

both its naphthalene oxygenase (NagGH) and its salicylate monooxygenase (S5H) [37]. Two dioxygenase enzymes (PhnI and PhnII) in *Sphingomonas* sp. CHY-1 utilize the same reductase (PhnA3) and ferredoxin (PhnA4) to catabolize a wide variety of PAHs, including chrysene and phenanthrene [38]. In *Staphylococcus aureus* separate ring cleavage enzymes (IsdG and IsdI), which also require molecular oxygen and an electron source, share a reductase (IruO) to cleave porphyrin rings associated with heme [39].

In other cases, the same electron transport proteins and oxygenase oxidize multiple compounds, much like the DD/DF oxygenase in RW1. For instance, the naphthalene dioxygenase (NDO) from *Pseudomonas* sp. 9816-4 utilizes the NAP reductase and NAP ferredoxin to oxidize naphthalene, indan, indene and (IS)-indanole [40].

Much like with the DD/DF oxygenase the benzoate oxygenase is located on the main chromosome of RW1, while *redA2* and

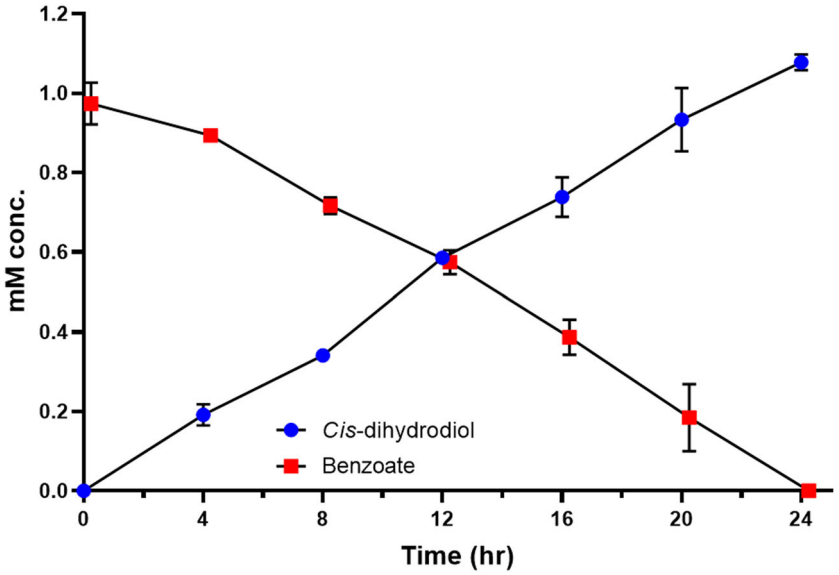



**FIGURE 7** | Conversion of benzoate to the *cis*-dihydrodiol by *Escherichia coli* BL21 DE3(pGGA_BenAB_RedA2_Fdx3) over a 24 h period. Data are averages from triplicates, and the error bars indicate their standard deviations.

*fdx3* are located on pSWIT02 [18, 20]. This underpins the sphingomonads tendency to have genes scattered across the genome that function together for a common catabolic pathway. It is very interesting that *redA2* and *fdx3* can function with the benzoate oxygenase since pSWIT02-like plasmids evolved under selective pressure of DF [41, 42] while benzoate degradation is much more common. This adds to the likelihood that other ferredoxins, most likely also on the chromosome, can work with either *redA1* or *redA2* to support the benzoate oxygenase. Since the known ferredoxins like *fdx3*, which incorporate 4 cysteine amino acids at the Fe-S core, did not give high yields of the *cis*-dihydrodiol there may be a novel 4 cysteine ferredoxin yet to be discovered. Alternatively, the benzoate oxygenase might also be able to use a ferredoxin that utilizes 2 cysteine and 2 histidine amino acids, however there is no evidence of an oxygenase which can use both classes of ferredoxins. Finally, there are no other genes necessary for benzoate degradation located on pSWIT02 as the dehydrogenase for the *cis*-dihydrodiol and the lower pathway genes for catechol degradation are on the chromosome.

We have functionally shown that *redA2* and *fdx3* work with the benzoate oxygenase to oxidize benzoate to the cis-dihydrodiol. This is the second oxygenase enzyme (in addition to DxnA1A2) and third known substrate (in addition to DD and DF) which the electron transfer components are implicated with. Although this can prove harmful to remediation efforts with RW1 due to the competition for electrons to complete the enzymatic reactions, it also opens the door for using these electron transfer components to augment other catabolic pathways. The plasticity of these enzymes warrants further experiments to see how many oxygenase enzymes they can support.

The benzoate dioxygenase in RW1 is a novel enzyme that does not show an identical sequence to other enzymes on the NCBI database. Previous expression experiments on benzoate dioxygenases have been performed in Pseudomonads, with *Pseudomonas putida* KT2440 [6] and *Pseudomonas arvilla* c-1 [43]. The benzoate dioxygenase in *Acinetobacter calcoaceticus* [14], *Rhodococcus* sp. strain 19070 [13], *Chromohalobacter* sp. strain HS-2 [44] and *Rhodococcus ruber* UKMP-5M [45] have also been analyzed. Here we functionally analyzed the second benzoate dioxygenase from the reclassified *Rhizorhabdus* genus, after a three component (methyl)benzoate dioxygenase that functions on benzoate in *Sphingomonas yanoikuyae* B1 [46].

### Author Contributions

**Igor Ivanovski:** conceptualization, investigation, writing – original draft, data curation, formal analysis, writing – review and editing, validation. **Suha Eleya:** conceptualization, investigation, data curation, formal analysis, validation. **Gerben J. Zylstra:** conceptualization, funding acquisition, supervision, formal analysis, writing – original draft, methodology, resources.

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### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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